Census Transformation and Application Modernization (CenTAM) Framework

Geospatial Blanket Purchase Agreement (BPA)

Scope and Task Areas

DRAFT

April 17, 2026

Table of Contents

1.0 Census Bureau Geospatial Initiatives Overview

The United States Census Bureau's (USCB) geospatial initiatives play a critical role in supporting its mission to produce accurate, timely, and relevant data about the nation's population, housing, and economy. Given the diversity of the US population, economic activities, and geographic areas, use of the latest and best geographic methodologies is critical to the USCB's ability to serve as the leading provider of statistical and geospatial data.

Geospatial capabilities are foundational to the USCB's operations, enabling precise mapping of addresses, boundaries, and demographic characteristics. These capabilities support key programs such as the Decennial Census, American Community Survey (ACS), and economic surveys, and are essential for ensuring geographic coverage, data quality, and operational efficiency.

In addition to supporting traditional geographic infrastructure and mapping activities, these capabilities are increasingly applied to modernize Economic Directorate programs and statistical methodologies. This includes establishment-level geocoding, integration of Business Register and survey data with spatial reference systems, spatial economic modeling, and the use of satellite imagery and computer vision models to supplement survey data. Recent experimental data products have combined traditional survey-collected data. Recent experimental data products have combined traditional survey-collected data with estimates derived from satellite imagery to produce hybrid monthly estimates of single-family housing starts, demonstrating the potential of geospatial AI to enhance economic indicators while maintaining statistical rigor.

USCB's objectives for its geospatial initiatives include the following:

- Developing criteria to define geographic areas
- Maintaining a large spatial database with spatial data for the United States, Puerto Rico, and the Island Areas
- Creating reference and thematic maps to support censuses and surveys
- Developing standards and manage governmental units, administrative, and statistical boundary data to meet obligations to the Federal Geographic Data Committee (FGDC)
- Digitizing spatial updates to the Master Address File/Topologically Integrated Geographic Encoding and Referencing (MAF/TIGER) system
- Processing and geocode millions of addresses
- Conducting research on geographic and address topics
- Developing programs to automatically enforce geographic rules, create products, geocode address lists, and create maps
- Creating tools to visualize geographic data

2.0 Geospatial BPA Scope

The scope of the Geospatial BPA is to provide application development, architecture engineering. Data science, and data management services supporting geospatial initiatives across the USCB. The support includes architects, engineers, data scientists, developers, testers, administrators and IT project management to enable the modernization of existing applications and the development of new applications.

Application development support encompasses a wide array of statistical applications utilizing various programming languages and technologies. These applications range from custom-built solutions to Commercial Off-The-Shelf (COTS) and Government Off-The-Shelf (GOTS) systems, as well as low-code, no-code and open-source based applications. These applications will range in the functionality they provide to the USCB including, but not limited to, applications for data collection, data ingestion, data processing, data cleaning, data validation, data aggregation, data storage, data modeling, data mining, data analytics and data science, reporting, business intelligence, content and data dissemination, geospatial data and all associated databases, data, applications, products, and services, geo-analytics, case management, process management, resource management, and workflow management.

Services provided under this BPA are further described within Task Areas 1-5.

Individual Call Orders will specify the support requirements, performance metrics, and deliverables applicable to Call Order performance.

These applications may support demographic, housing, and economic programs, including survey production systems, establishment-based data systems, business frame integration, hybrid estimation methodologies, and spatial analytics supporting official and experimental data products.

3.0 Task Areas

The Contractor shall develop solutions and/or provide the capabilities and support in the following task areas to enable geospatial and spatially enabled initiatives across decennial, demographic, economic, and experimental mission portfolios.

- Task Area 1 – Order and Program Management
- Task Area 2 – IT Project and Program Management
- Task Area 3 – Architecture and Engineering
- Task Area 4 – Application Development and Management Services
- Task Area 5 – Advanced Data Science and Data Management

3.1 Task Area 1 –Order and Program Management

Task Area 1 requires services focused on providing comprehensive, continuous contract and program management oversight to ensure effective governance, coordination, reporting, quality, compliance, and performance across all orders issued under this BPA.

For awarded Call Orders, the Contractor shall:

1. Manage all contract and program activities in alignment with the latest publication of the Project Management Institute’s (PMI) Project Management Body of Knowledge (PMBOK).
2. Establish, staff, and operate a Contract Program Management Office (CPMO) that provides centralized program and contract oversight, serves as the focal point for coordination with the Government and ensures compliance with all contractual requirements.
3. Designate a Program Manager, with a designated backup, as the authoritative point of contact for all contract-level communications, maintaining full situational awareness of program activities.

4. Develop, document, maintain, and adhere to the following management plans as required:
    a) Contract Management Plan (CMP)
    b) Staffing Plan
    c) Subcontractor Management Plan
    d) Quality Control Plan (QCP)
    e) Transition-In Plan (TIP)
    f) Transition-Out Plan (TOP)
5. Recruit, onboard, supervise, retain, and offboard qualified personnel—prime and subcontractor—ensuring all personnel meet suitability, security, certification, and training requirements prior to assuming duties.
6. Manage staffing continuity, surge staffing needs, personnel transitions, and subcontractor performance, ensuring alignment with Government requirements and maintaining complete and accurate personnel documentation.
7. Maintain active risk management processes, including identification, assessment, tracking, mitigation, and reporting of contract-level and program-level risks and issues.
8. Establish and maintain communication protocols, meeting cadences, escalation processes, documentation standards, and communication logs to support effective collaboration with the Government.
9. Develop and maintain a contract level schedule baseline, a Contractor Work Breakdown Structure (CWBS) aligned with the Government's DWBS, and provide timely, accurate schedule inputs to support the Decennial Integrated Master Schedule (IMS).
10. Provide data call and audit support for internal and external audits, inquiries, and oversight requests.
11. Maintain a Lessons Learned process and log, documenting insights, improvement opportunities, and corrective actions, and incorporate lessons learned into updates of management artifacts.
12. Execute transition-in and transition-out activities in accordance with Government-approved TIP and TOP, ensuring continuity of operations, complete transfer of knowledge and artifacts, return of Government assets, and timely completion of all close-out activities.
13. Support and conduct required meetings and deliver required reports to keep the Government informed on status, cost, schedule, staffing, risks, performance, quality, and contractual compliance.
14. Support baseline development and reviews, including Performance Measurement Baselines (PMBs) and Integrated Baseline Reviews (IBRs), and maintain traceability to DWBS elements.
15. Manage all deliverables in accordance with Section F and maintain a deliverables tracker aligned to CWBS, DWBS, IMS, and associated milestones.

16. Ensure compliance with all applicable laws, regulations, directives, and standards, including but not limited to security, privacy, accessibility (Section 508), and Census-specific policies, and notify the Government of any compliance issues.
17. Collaborate with other Government and contractor teams as directed, sharing information, artifacts, and resources necessary for integrated execution while avoiding conflicts of interest.

3.2 Task Area 2 – IT Project and Program Management

Task Area 2 requires services for the planning, tracking, management, and deployment of USCB IT programs, projects, and initiatives including, but not limited to, program planning, governance, training, organizational change management, and product delivery management.

3.2.1 IT Project and Program Management Support

The Contractor shall:

1. Provide program management and planning to support the coordination and execution of USCB IT programs, projects, and activities, using agile and other development methodologies.
2. Serve as a member and/or provide IT project management support to product teams. These product teams at the USCB include but are not limited to: Value Teams, Operational Delivery Teams, and Integrated Product Teams (IPTs).
3. Develop, deliver, maintain, provide status on, and manage project schedules across programs.
4. Support archiving of System Development Life Cycle (SDLC) deliverables.
5. Collect, analyze, and report metrics that provide insight necessary to accurately predict and effectively manage the cost, schedule, technical performance, and risks.
6. Conduct and document periodic lessons learned with recommendations for improvement.
7. Provide risk management support services for the USCB's program risk, issue, and opportunity management activities, including, but not limited to, identifying, reporting, tracking, and resolving risks and supporting program/project risk and issue boards.
8. Provide change control support services for the USCB's project and program-level change control activities.
9. Support the design, development, and implementation of change and configuration management processes, standards, and templates.
10. Support the development and update of IT strategic plans, program governance, and alternative analysis for programs and projects in the USCB.
11. Develop, document, implement, and maintain technical documents, including, but not limited to, systems engineering, architecture, workflow control, software, testing, and system operations.
12. Develop, document, implement, and maintain communication plans and coordinate efforts with other Census Contractors, stakeholders, and other organizations as directed.

3.2.2 Training

The Contractor shall:

1. Perform a current state assessment of training needs to meet USCB transformation and application modernization needs.
2. Develop, document, implement, and maintain training plans, manuals and other documentation or aids. Training topics include but are not limited to topics such as application development, data science, cloud adoption, agile practices, requirements development, security, business process modeling, change management, and configuration management.
3. Track training according to the training plan.
4. Identify and provide any additional training and written documentation including, but not limited to, hands-on exercises, other tools training, demonstrations, seminars, etc. required by end-users, technicians, or any other staff for implementation, maintenance, and use of applications, work products, and deliverables developed by the Contractor as part of this contract.

3.2.3 Organizational Change Management (OCM)

The Contractor shall:

1. Develop, document, implement, and maintain knowledge transfer strategies and OCM plans that will result in minimal costs and maximum long-term adoption of USCB Programs' transformation and application modernization initiatives that align with USCB objectives.
2. Develop, document, implement, and maintain a communication plan to support technology transformation and application modernization for the USCB.
3. Facilitate workshops to support technology transformation initiatives.
4. Design, develop, and execute an integrated approach to knowledge management/enterprise information management.
5. Develop, document, implement, and maintain the development, maintenance, and implementation of a training and organizational change management repository.
6. Collect and report periodic lessons learned including, but not limited to, organizational change management metrics.
7. Develop, document, implement, and support change management configurations and methodologies for the USCB.

3.2.4 Product Delivery Management

The Contractor shall:

1. Provide delivery management services including, but not limited to, IT and non-IT product delivery management.
2. Provide support in the planning, tracking, logistics, management, and deployment of various components, systems, and processes.

3. Monitor and manage integration points, interdependencies, and milestones among the USCB service/solution providers for product delivery and collaborate with stakeholders and service/solution providers to resolve blockers.
4. Coordinate the implementation of the artifact and phase gate review processes for applications as they progress through the SDLC.
5. Support phase gate program reviews including, but not limited to, preparing review materials, facilitating review sessions, capturing, and tracking of action items, and producing summary reports with metrics.
6. Develop and update runbooks including but not limited to providing a comprehensive set of steps detailing system and user interactions when releasing new software or features into a production environment.

3.2.5 System Level Testing

The Contractor shall:

1. Provide system-level software testing focused on validating the holistic functionality, performance, and security of the entire system or program ecosystem, including but not limited to performance testing, scalability testing, integration testing, exception testing, pairwise and security testing
2. Develop and perform automated testing services that support system-wide validation across integrated environments.
3. Develop an integrated, system-wide end-to-end test management approach that supports requirements traceability, defect analysis, and retrospectives throughout the software delivery pipeline.
4. Develop, update, maintain, and disseminate System Test Analysis Reports that reflect testing outcomes across the entire system or program ecosystem.
5. Collaborate with system and/ or component owners on defect resolution including, but not limited to, categorizing, tracking, and dispositioning all software defects.
6. Provide system test execution support through manual and automated testing methods across integrated platforms.
7. Create and maintain system-level business threads and workflows to support the testing activities.
8. Develop, update, maintain, and disseminate a System Defect Management Plan.
9. Develop system test plans including, but not limited to, system testing, automation, end-to-end testing, integration testing, data validation, system integration, output testing, and exception testing across multiple components.
10. Create integrated test data to support system-level testing across multiple applications, services, and platforms.
11. Conduct multi-layer testing (including the validation of outputs) across various platforms (e.g., smartphones, laptops, tablets, and desktops) to validate consistent system behavior.

12. Conduct system testing in the fully integrated program ecosystem environment, validating end-to-end functionality, data flow, and performance across all relevant systems and external interfaces.
13. Support and participate in reviews such as test readiness and test analysis reviews.
14. Develop and maintain system-level test cases, test scripts, test scenarios, and test metrics that reflect cross-system requirements.
15. Maintain source code and documentation repositories relevant to system level integration and testing activities.
16. Coordinate system-level testing activities across teams, environments, and stakeholders to provide alignment, scheduling, and resource availability for integrated test efforts.

3.3 Task Area 3 – Architecture and Engineering

Task Area 3 requires services for systems architecture and engineering inclusive of all phases of the systems' development and operational lifecycle. This includes but is not limited to understanding the USCB architectural landscape, defining, and documenting requirements, and generating and validating the design with the entire product cycle in mind – operations, performance, testing, cost and scheduling, training and support, security, and disposal.

3.3.1 Systems Architecture Engineering

The Contractor shall:

1. Conduct a comprehensive review and assessment of current systems architecture and engineering footprints, plans, standards, roadmaps, schedules, requirements, and other planning artifacts.
2. Develop, document, implement, and maintain a system architecture and engineering roadmap, ensuring alignment with on-prem, SaaS, and cloud/multi-cloud architectures, practices, technologies, tools, services, and capabilities.
3. Develop and document the gap analysis findings and provide recommendations.
4. Implement, maintain, and support the approved recommendations of the gap analysis findings.
5. Conduct requirements analysis; develop a comprehensive systems design blueprint; utilize modeling and simulation to predict system behavior and performance; integrate system components in compliance with architecture; perform verification and validation to confirm system architecture meets all requirements.
6. Develop, document, implement, support, and maintain systems engineering plans including, but not limited to, Systems Engineering Management Plan and Systems Engineering Plan.
7. Develop, maintain, and provide periodic reports, status updates, and data visualizations regarding system architecture and engineering status and activities.
8. Develop and maintain application and database inventories for the systems within the USCB.

9. Collaborate with internal and external service providers, solution providers and third-party vendors to support implementation of the systems architecture and engineering processes and procedures.
10. Plan, coordinate, execute, and/or participate in ongoing architecture and engineering reviews of systems, programs, infrastructure, and supporting services/capabilities to verify that they are properly engineered and developed in a fashion that is in alignment with architecture and engineering standards and guidelines, and provide best practice recommendations as appropriate.

3.3.2 Cloud Architecture and Engineering

The Contractor shall:

1. Assist with the development, management, and support of cloud control planes to provide management across cloud application deployment.
2. Develop, document, implement, and maintain cloud architecture and engineering plans, including, but not limited to tagging strategies and standards, cloud messaging standards, and cloud development standards.
3. Review and analyze cloud security tools and implementations and make recommendations based on Census Enterprise security requirements and best practices.
4. Establish and maintain a roadmap of required cloud services, capabilities, and requirements based on USCB, security, and operational milestones and requirements, and make recommendations for development, implementation, enhancement, and maintenance.

3.3.3 Systems Engineering Management

The Contractor shall:

1. Assist in creating and maintaining systems engineering management process documentation including, but not limited to, Defect Management Plans, Disposition Management Plans, and Operational Delivery Management Plans.
2. Support technical integration within all functional areas to provide consistent systems engineering processes and the execution of those processes.
3. Communicate and socialize the updates made to the systems engineering processes with all impacted stakeholders and manage the transition to, and adoption of, the updated processes.
4. Assist with the creation and maintenance of IT planning and IT strategy documents for the USCB such as the USCB IT Strategy and Roadmap.

3.3.4 Requirements Management

The Contractor shall:

1. Assist in eliciting, collecting, documenting, managing, facilitating, analyzing, validating, tracing, and allocating USCB requirements.

2. Assist in supporting the development and maintenance of a requirements repository.
3. Assist in the creation and maintenance of a capabilities/features catalogue.
4. Assist in the creation and maintenance of a requirements management plan.
5. Collaborate with stakeholders, subject matter experts, and service solution providers to design and implement innovative, efficient, and sustainable solutions that meet project requirements, enhance functionality, and ensure compliance with industry standards and regulations.
6. Develop, document, implement, support, and maintain the enterprise architecture framework and plans including, but not limited to:
    a) Business Architecture
    b) Logical Architecture
    c) Enterprise Architecture
    d) System of Systems Architecture
    e) Solution Architecture
    f) Information Architecture
    g) Application Architecture
    h) Data Architecture
    i) Security architecture

3.3.5 Facilitation Services

The Contractor shall:

1. Organize, design, lead, and participate in meetings, working sessions, troubleshooting sessions, and other collaborative sessions with solution/service providers, federal and contract staff, third parties, and other stakeholders as necessary.
2. Capture and/or scribe session meeting notes, decisions/recommendations, and action items.
3. Track action items to resolution and escalate any blockers.

3.4 Task Area 4 – Application Development and Management Services

Task Area 4 requires the full spectrum of services for application design and development, architecture, testing, security, operations and maintenance (O&M), cloud application migration, research and development (R&D), detailed documentation, and innovation.

3.4.1 Application Design and Development

The Contractor shall:

1. Provide the full range of software engineering services to enhance, maintain, and/or refactor existing applications and design and develop new applications.
2. Provide software engineering services using USCB and Commerce Department software development lifecycle frameworks and guides, as well as industry best practices and methodologies such as, but not limited to Agile, DevOps and DevSecOps.
3. Document and formalize industry best practices to be incorporated and/or adopted as USCB methodologies.
4. Provide modern software engineering approaches that enhance speed, security, and value of applications and service delivery including, but not limited to, approaches for emerging trends and technologies such as platform engineering, infrastructure as code/everything as code, and the use of generative AI for code development and/or testing.
5. Incorporate functional and non-functional requirements in designs.

3.4.2 Application Architecture

The Contractor shall:

1. Provide solution/application-level architecture support in coordination with the USCB's enterprise architecture.
2. Develop, document, and maintain design patterns including, but not limited to, service levels, service-oriented architecture, and application programming interfaces (APIs).
3. Develop and maintain Concepts of Operations based on business requirements by application.
4. Identify, document, and assist the government in the development of microservice specifications aligned to overall USCB vision for service-oriented architecture.
5. Provide best practices and solutions that meet USCB functional and non-functional requirements to help migrate the USCB application portfolio into a cloud-based, service/mesh-oriented architecture, comprised of micro/mini services and coarse/granular APIs.
6. Assist in troubleshooting or identifying solutions to reduce costs and/or improve performance for compute, compute functions, high performance computing, storage, or other services.
7. Assess program needs to determine whether new applications need to be achieved through GOTS, COTS or other application types.

3.4.3 Application Testing

The Contractor shall:

1. Provide application-level software testing focused on verifying the functional and non-functional requirements of individual applications including, but not limited to, performance, scalability, integration, exception handling, pairwise testing and security.
2. Develop and perform automated testing services tailored to the application's architecture and business logic.

3. Develop and implement a test management approach that supports requirements traceability, defect analysis, and retrospectives throughout the software delivery pipeline.
4. Develop, update, maintain, and disseminate Application Test Analysis Reports that reflect testing outcomes specific to the application.
5. Collaborate with application developers on defect resolution including, but not limited to, categorizing, tracking, and dispositioning all software defects within the application scope.
6. Provides application and database test execution support through manual and automated testing.
7. Create and maintain application-level business threads and workflows to support the application testing activities.
8. Develop, update, maintain, and disseminate a Defect Management Plan.
9. Develop application test plans including, but not limited to, application testing, automation, workflow testing across all internal components and immediate dependencies of the application, integration testing with defined components or interfaces, data validation, output testing, and exception testing.
10. Create application-specific test data that supports validation of business rules, data models, and user scenarios.
11. Conduct multi-layer testing (including the validation of outputs) across supported platforms (e.g., smartphones, laptops, tablets, and desktops) to ensure consistent application behavior.
12. Conduct application testing in the integrated environment specific to the individual application, validating functionality and performance with its immediate internal and external dependencies (e.g. databases, APIs, other services).
13. Support and participate in reviews such as test readiness and test analysis reviews.
14. Develop and maintain application-level test cases, test scripts, test scenarios, and test metrics aligned with application requirements.
15. Maintain application specific source code and documentation repositories relevant to testing activities.
16. Facilitate the release of software from development to testing, staging, user acceptance testing (UAT), and production environments.
17. Coordinate application-level testing activities with development teams, QA resources, and stakeholders to ensure timely execution, environment readiness and alignment with release schedules.

3.4.4 Application Security

The Contractor shall:

1. Provide application development and management services in accordance with all federal IT security guidance, policies, and regulations applicable to the Commerce Department and the USCB including, but not limited to, Zero Trust and the USCB Risk Management Framework.

2. Develop and implement specialized security solutions, toolsets, toolkits, frameworks, threat modeling assessments, vulnerability assessments, guides, and processes.
3. Support development teams with security testing, to include test plan development, and security testing services, and documentation/reporting for USCB applications.
4. Develop, document, implement, and maintain an application security program, identify stakeholders, define security requirements, and establish metrics and reporting to track progress and improvement over time.
5. Support the development and management of security related documentation, including, but not limited to, Interagency Security Agreements, Software Bill of Materials, standards for systems, application, and data, and playbooks related to secure coding, security integration, security operations, and incident response.
6. Scan application components to detect outdated and/or vulnerable libraries and license issues in third-party libraries.
7. Train and educate developers and testers on security best practices and identifying common security vulnerabilities found in software and applications.
8. Develop and implement best practices to include, but not limited to secure coding practices, secure session management, secure file handling practices, secure communications protocols and encryptions, secure error handling, exception management, logging, and monitoring mechanisms.
9. Respond to security-related data calls from, including, but not limited to, Office of Information Systems, and Federal Information Security Modernization Act (FISMA).
10. Consult with stakeholders, such as procurement and legal teams, to include appropriate security clauses and requirements in contractual agreements with third-party vendors.
11. Assist in the evaluation and selection of new tools and applications and provide written documentation of assessments, findings, and recommendations.
12. Implement mechanisms for continuous monitoring and assessment of third-party software components to ensure ongoing compliance with security requirements and timely identification of any potential vulnerabilities or risks.
13. Develop and implement secure software distribution and update mechanisms, including secure repositories and digital signatures, to ensure the integrity and authenticity of software components to include supply chain attestation.
14. Support Continuity of Operation Plans (COOP), Information System Contingency Plans and Disaster Recovery (DR) that address software continuity and comply with National Institute of Standards and Technology guidelines and the Bureau’s IT Security program.
15. Develop, document, implement, and maintain COOP and DR strategies, architectures, as well as designing and conducting tabletop exercises and simulations to validate the effectiveness of COOP and disaster recovery plans.
16. Provide assessments of these tabletop activities, document recommendations, and perform necessary training for stakeholders based on these recommendations.

3.4.5 Application Operations and Maintenance

The Contractor shall:

1. Handle release management by planning and coordinating releases, ensuring smooth transitions to all environments, documenting processes, monitoring progress, and conducting post-release reviews while managing risks, ensuring compliance with policies.
2. Develop and maintain systems architecture and engineering artifacts, in alignment with USCB Enterprise Architecture.
3. Collaborate with IT service management and enterprise infrastructure teams to monitor application performance.
4. Provide operations and maintenance (O&M) support of applications in all environments including, but not limited to, maintenance requests such as patching, defect resolution, minor enhancements, development workflow and providing updated information to the appropriate stakeholders.
5. Collaborate, participate, and be available for rapid response teams, and participate in Root Cause Analysis and troubleshooting to assess and diagnose problems, identify root causes of technical problems, and recommend solutions.
6. Establish and implement procedures, metrics, and documentation on defect management and technical debt reduction.
7. Collaborate with USCB stakeholders to minimize impacts to the mission due to planned disruptions due to system updates including, but not limited to, change management policies and procedures.
8. Maintain, implement, and/or operate backup and recovery plans in collaboration with system, application, and infrastructure stakeholders.
9. Assist in performing cost analysis including, but not limited to, total cost of ownership costs, infrastructure and/or cloud costs, and other O&M costs.
10. Create and maintain custom reports, graphics, and dashboards to support application operations and monitoring.
11. Provide hardware infrastructure support and services.
12. Provide administration support for COTS, GOTS, open-source, SaaS and in-house developed software.
13. Conduct reviews, audits, and risk analysis and provide findings and recommendations.
14. Provide operations support for applications including, but not limited to, Tier 3 support. The Contractor must track support activities and metrics to be used by the government to assess service level quality.

3.4.6 Cloud Application Migration

The Contractor shall:

1. Support the USCB in articulating a cloud application migration value proposition defined for both business and IT.
2. Establish, implement, and maintain a cloud application migration strategy and roadmap that aligns to USCB IT strategic goals, enterprise architecture roadmap, and security guidelines.

3. Establish, implement, and maintain decision principles for migration, informed by application and team readiness, business priorities, application modernization strategies, and vendor capabilities.
4. Establish, implement, and maintain an application rationalization/optimization plan and migrate applications to the cloud.
5. Develop, document, implement, and maintain a cloud adoption framework.
6. Develop, document, implement, and maintain a cloud application governance process.
7. Establish, implement, and maintain the appropriate cloud footprint optimization and evolution capabilities for ongoing FinOps, DevSecOps, and Zero Trust Model disciplines for cloud application migration.

3.4.7 Research, Development, and Innovation

The Contractor shall:

1. Perform evaluations and/or analysis of alternatives of new or emerging technology applications for requirements including, but not limited to, evaluations such as technical feasibility, operational feasibility, cost, scalability, security, and risks.
2. Design, develop, document, and implement application prototypes and pilots to determine feasibility of technologies, emerging, or otherwise. The development and/or configuration of pre-production applications, COTS/GOTS, or open source-based prototypes may be required.
3. Assist in the evaluation of the current environment and make recommendations on a research environment that will allow for the analysis of research and development activities.
4. Perform engineering studies and analyses such as, including, but not limited to, demonstrating the feasibility of conceptual approaches, evaluating design risk, evaluating the viability of potential solutions, alternatives to various technical issues and challenges, emerging products, or technology, and comparing and analyzing trade-offs.
5. Perform technology related studies and analyses including, but not limited to, logistics, supportability, engineering, financial, operational, business processes, and applications to include mobile applications, modernization of existing systems and applications, interoperability, and information sharing.

3.5 Task Area 5 – Advanced Data Science and Data Management

Task Area 5 requires services for data management, analytics, reporting, and advanced analytics/data science to support the USCB with foundational needs and innovation objectives. This includes the use, creation, transformation, and/or analysis of the agency's critical data assets for mission purposes.

3.5.1 Advanced Data Science

The Contractor shall:

1. Develop and implement advanced algorithms, models, and frameworks that leverage the latest advances in technologies including, but not limited to, machine learning, deep learning, neural networks, natural language processing, Generative AI (GAI), large

language models, pattern recognition, geospatial analytics, data wrangling, algorithmic programming, digital twin development, and predictive modeling.

2. Leverage automation and machine learning as part of integrated IT solutions to manage data, predict scenarios, and make recommendations.
3. Develop, document, and test prototypes and proofs of concepts leveraging advanced data science.
4. Assist the USCB with moving advanced analytics and data science pilots into production and support maintenance of those capabilities.
5. Create, maintain, and use synthetic data as applicable.
6. Follow AI ethical principles in accordance with industry and Federal Government best practices, to closely manage AI risk and ensure trustworthiness of resulting models and other advanced data science products that use AI/ML.
7. Adhere to all policies, guidance, regulations, and directives on the use of AI within the USCB.
8. Provide expertise and services focused on AI explainability to USCB communicating, potential uses of AI to its leadership, oversight bodies, and the public.
9. Perform interactive data analysis in support of USCB data quality.

3.5.2 Data Management

The Contractor shall:

1. Develop, document, implement, and maintain data management work products and/or capabilities using secure methodologies to meet business requirements and functional specifications while maintaining data integrity and lineage and governance. Services may support demographic, housing, and economic mission portfolios, including longitudinal economic data sets, establishment –based systems, and spatially-enabled administrative record integration.
2. Support may include database management systems, data integration tools, data quality tools, master data management platforms, address data products, geospatial data products, cartographic services, map products, data warehouses, data lakes, hybrid geospatial-economic data environments, and data pipelines.
3. Design, develop, and support applications, systems, databases, data stores, and repositories used to access manipulate, manage, store, and back up data required to support all USCB operations.
4. Assess develop, update, and maintain data integration workflows (including real-time) that include data from various sources such as databases, applications, APIs, and file formats, imagery services, and external data providers.
5. Define and automate Extract, Load and Transform (ELT/ETL) operations ensuring scalability, traceability, reproduceability and performance.
6. Enhance, update, maintain, and/or develop data dictionaries, catalogues, and repositories that align with enterprise data standards, and document authoritative data sources in these and other artifacts as appropriate.

7. Services may include ingestion, storage, processing, and management of large-scale geospatial datasets, including satellite imagery, raster and vector datasets, spatial time-series data, and computer vision model outputs, and integration of those outputs with survey product systems and Business Register systems.
8. Provide support for defining and implementing conceptual, logical, and physical data modeling concepts, techniques, and tools.
9. Develop, document, implement, and maintain efficient and scalable database/schema designs (across various database management systems) that meets requirements and available data management/governance standards.
10. Research, develop, document, implement, and maintain metadata management solution(s) that capture, store, and manage metadata from various sources such as databases, applications, and data integration processes that support a wide range of metadata types including technical, business, and operational metadata.
11. Provide capabilities for tracking, documenting, and identifying issues, anomalies, and inefficiencies across the lineage of data elements in various sources, systems, and/or applications including their origin, transformations, and usage across different systems and processes. These capabilities include but are not limited to data governance platforms, graph analytics, and AI/ML solutions to investigate, track, and/or remediate data lineage issues.
12. Support USCB in data profiling, data quality, and data integrity support services.
13. Assist with the creation, establishment, and publication of data quality and integrity standards, policies, metrics, and frameworks as needed.
14. Provide support for data currency policies, processes, and/or procedures to align timeliness and recency of data sharing, integration, and other processes with business needs.
15. Design, coordinate, and manage data archiving plans and strategies.
16. Support and document archival, transfer, migration and/or disposition of data.
17. Provide support to develop, maintain, and/or provide input on data management strategies, policies, and standards.
18. Support the creation and maintenance of tagging strategies and standards to help manage, search for, and filter resources as well as maintain cost traceability.
19. Provide expertise for record linkage including using-in place systems and methodology, authoring new algorithms, and applying methods and systems to existing problems.
20. Support database software installations and upgrades include patching and upgrading the database software as per the USCB mandates and stakeholder schedules.
21. Participate in after-hours upgrades, maintenance, database support, troubleshooting, and on-call availability weekend and evening activities (24/7) as needed and coordinating service calls when required by Contracting Officer’s Representative (COR).

3.5.3 Business Intelligence, Reporting, and Analytics

The Contractor shall:

1. Provide data visualization support, workflows, models, and dashboards.
2. Develop reproducible reports and analyses, such as benchmarking, for performance and efficiency.
3. Manage and govern analytics and reporting capabilities leveraging applicable and available data access, quality, and integrity policies, standards, and processes.
4. Support development, enhancements, and maintenance of data reporting and data discovery models and frameworks for USCB Programs.
5. Support integration of data sources, storage, and other tools/applications with available data visualization tools to create interactive reports and dashboards.
6. Create and customize reporting templates to meet the specific needs of the organization, including report layouts, formatting options, and data visualization elements such as charts, graphs, and tables.
7. Ensure analytics and reporting capabilities are managed and governed by applicable and available data access, quality, and integrity policies, standards, and processes.

3.5.4 Model Development

The Contractor shall:

1. Develop, train, explain, validate and maintain models including, but not limited to, AI/ML models, computer vision models, hybrid survey-model estimation frameworks, small-area estimation models, architecture/engineering models, application models, and testing models.

   Develop and maintain business process models, interface data flow documentation (e.g., data exchanges and system interconnections), integrated operations diagrams, and functional and technical diagrams.